

Climate 5000 Series - Gen 4

24K BTU Single Zone Max Performance System

Wall Mounted IDU

Engineering
Submittal
Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

24K Btu Single Zone Max Performance System - Wall Mounted IDU

Indoor part number	8-733-962-675
Indoor model number	BMS500-AAU024-1AHWXD
Outdoor part number	8-733-962-691
Outdoor model number	BMS500-AAS024-1CSXHD

Efficiency Data		AHRI 210/240 2023
SEER2		21.0
EER2		13.0
HSPF2 Region IV		13.5
HSPF2 Region V		9.8
COP(47F)		3.42
COP (5F)		2.34

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	12600~27900
	Rated Capacity	24000
	Current	8.3
	Input Power	1846
Heating	Capacity Range	11800~29700
	Rated Capacity (47F)	29000
	Rated Capacity (5F)	24200
	Current	11.0
	Input Power	2485

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 24.9
Max. fuse indoor/outdoor (A)	15.0 / 25.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 620
Input (W)	805 / 2045
Rated current (RLA/A)	4.15 / 9.30

Operation Range		
ODU	Cooling (°F / °C)	-22~122 / -30~50
	Heating (°F / °C)	-22~75 / -30~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

Piping	
Refrigerant precharge (oz/g)	63.49 / 1800.00
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	3/8" / Ø9.52
Gas side (in/mm)	5/8" / Ø15.9
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.32 / 30
Max. refrigerant pipe length (ft/m)	164.00 / 50
Max. difference in height (ft/m)	82.00 / 25

System Data		
Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		5.7
IDU	Fan motor model	ZKFP-58-8-1-10
	Fan motor qty (ea)	1
	Fan motor input (W)	58.0
	Fan speed (RPM)	1000 / 850 / 630
	Air flow, Turbo / H / M / L / Si (CFM)	812.3 / 618.0 / 412.0 / 317.8 / /
	Approx. Air Throw Range ft (m)	29.5 / 9
	Noise level, Turbo / H / M / L / Si (dB(A))	54.5 / 43.5 / 38.5 / 27 / 16
	Net weight (lbs / kg)	43.65 / 19.8
	Gross weight (lbs / kg)	55.56 / 25.2
ODU	Fan motor model	ZKFN-120-8-2
	Fan motor qty (ea)	1
	Fan motor input (W)	150
	Fan motor speed, H / M / L (RPM)	950 / 800 / 500
	Air flow, Max (CFM)	2235.3
	Noise level, Max (dB(A))	61.5
	Net weight (lbs / kg)	130.29 / 59.1
	Gross weight (lbs / kg)	141.31 / 64.1

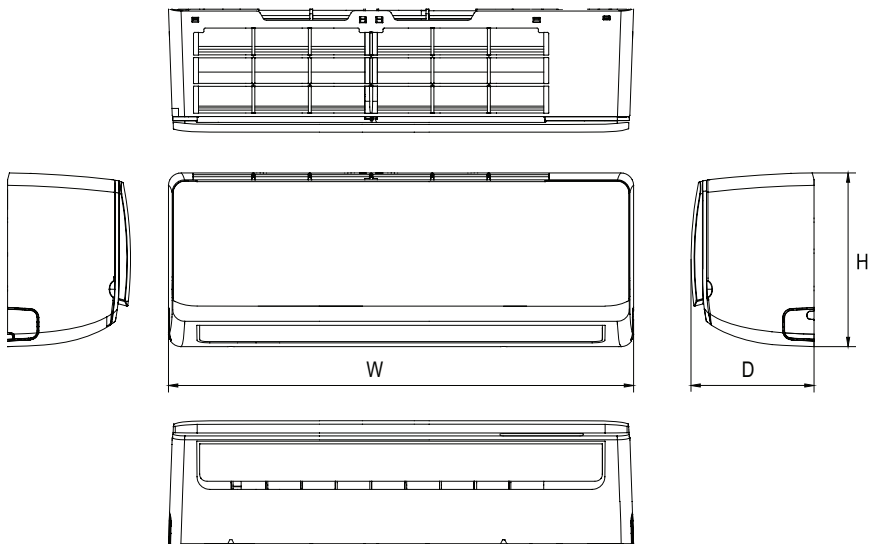
Accessories	
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK3-7X1 (8733951015)	25ft Lineset

Standard Parts	
8-733-969-088	Remote Control

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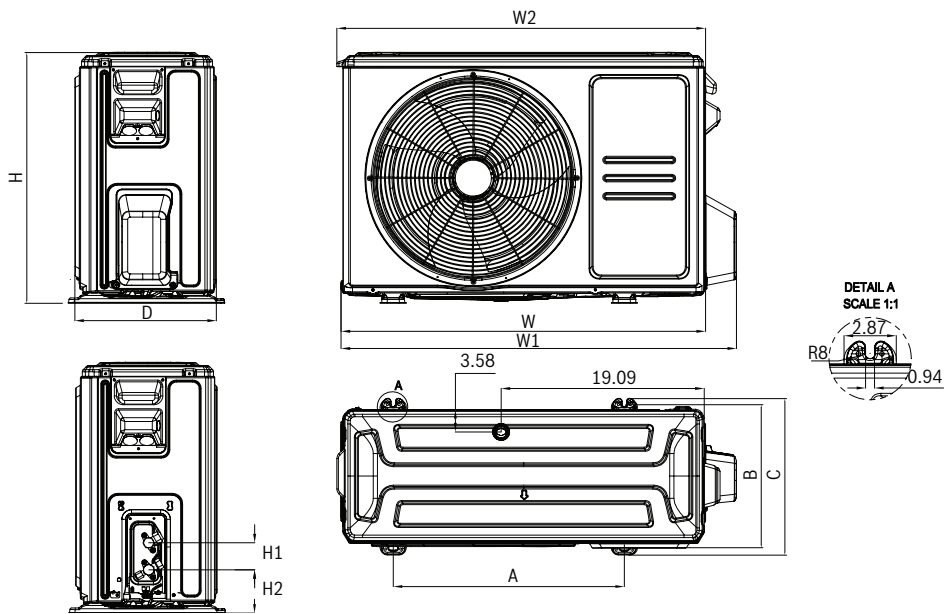


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU024-1AHWXD	44.88	10.83	14.57

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS024-1CSXHD	37.24	16.14	31.89	40.55	37.24	2.36	4.25	26.5	15.87	17.9



Cooling Performance - Indoor Dry Bulb Temperature at 718 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	20056	20047	20333	20417	20979	20926	20908	20895	20893	20876	21551	21524	21504	21501	21480	22732	22726	22680	22671	22623
	S/T	0.80	0.91	0.99	0.99	0.99	0.64	0.75	0.84	0.86	0.97	0.55	0.65	0.74	0.75	0.86	0.39	0.47	0.55	0.57	0.67
	PI	1.32	1.32	1.32	1.32	1.31	1.31	1.31	1.31	1.31	1.32	1.31	1.31	1.31	1.31	1.31	1.32	1.32	1.32	1.32	1.32
-13°F (-25.0°C)	TC	21727	21717	22027	22119	22727	22669	22651	22636	22634	22616	23347	23318	23296	23293	23270	24627	24620	24570	24560	24508
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	1.37	1.37	1.37	1.37	1.36	1.36	1.36	1.36	1.36	1.36	1.36	1.36	1.36	1.36	1.36	1.37	1.37	1.37	1.37	1.37
-4°F (-20.0°C)	TC	23399	23388	23722	23820	24476	24413	24393	24378	24375	24355	25143	25112	25088	25084	25060	26521	26514	26460	26450	26394
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	1.42	1.42	1.42	1.42	1.41	1.41	1.41	1.41	1.41	1.41	1.41	1.41	1.41	1.41	1.41	1.42	1.42	1.42	1.42	1.42
0°F (-17.8°C)	TC	24175	24161	24609	24712	25391	25212	25184	25165	25162	25415	25953	25916	25884	25880	25868	27349	27343	27340	27344	27362
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.43	1.45	1.45	1.45	1.45	1.45
5°F (-15.0°C)	TC	24868	24954	25524	25642	26432	26035	26041	26043	26070	26458	26867	26874	26880	26880	26916	28443	28435	28449	28453	28465
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	1.44	1.44	1.44	1.45	1.45	1.45	1.45	1.45	1.45	1.45	1.46	1.46	1.46	1.46	1.46	1.49	1.49	1.49	1.49	1.49
14°F (-10.0°C)	TC	27375	27361	27920	28051	28920	28734	28709	28689	28686	28955	29719	29681	29651	29646	29617	31600	31590	31530	31519	31452
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	1.06	1.06	1.06	1.06	1.06	1.05	1.06	1.06	1.06	1.06	1.05	1.05	1.05	1.05	1.06	1.06	1.06	1.06	1.06	1.06
17°F (-8.3°C)	TC	28142	28234	28886	29027	29970	29543	29555	29561	29585	30002	30551	30563	30575	30576	30614	32492	32483	32478	32484	32511
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.93	0.93	0.93	0.93	0.93
23°F (-5.0°C)	TC	25883	25995	26667	26799	27682	27194	27202	27205	27241	27712	28132	28142	28149	28150	28195	29913	29904	29925	29929	29946
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.92	0.92	0.92	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.95	0.95	0.95	0.95	0.95
35°F (1.7°C)	TC	21582	21582	21709	21770	22472	22590	22589	22591	22592	22652	23298	23296	23297	23298	23300	24609	24603	24602	24601	24603
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.87	0.87	0.88	0.88	0.91	0.92	0.92	0.92	0.92	0.92	0.96	0.96	0.96	0.96	0.96	1.05	1.04	1.04	1.04	1.04
41°F (5.0°C)	TC	25843	25843	26049	26230	27005	27045	27044	27045	27045	27140	27888	27886	27888	27888	27890	29453	29446	29444	29443	29446
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	1.14	1.14	1.15	1.16	1.20	1.21	1.21	1.21	1.21	1.21	1.26	1.26	1.26	1.26	1.26	1.38	1.37	1.37	1.37	1.37
47°F (8.3°C)	TC	30105	30125	30529	30666	31573	31501	31500	31501	31500	31646	32479	32477	32479	32478	32478	34298	34289	34286	34285	34288
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	1.40	1.40	1.43	1.44	1.50	1.49	1.49	1.49	1.49	1.50	1.56	1.56	1.56	1.56	1.56	1.70	1.70	1.70	1.70	1.70
50°F (10.0°C)	TC	29226	30031	31044	31224	32435	30876	30893	31107	31205	32466	32081	32123	32140	32141	32450	34292	34383	34471	34482	34525
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	1.60	1.60	1.60	1.60	1.59	1.60	1.60	1.60	1.60	1.59	1.59	1.59	1.59	1.59	1.59	1.59	1.59	1.59	1.59	1.59
59°F (15.0°C)	TC	28211	29200	30237	30421	31658	29848	29889	30228	30453	31690	31079	31111	31122	31167	31716	33373	33440	33507	33514	33542
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	1.68	1.68	1.68	1.69	1.69	1.68	1.68	1.68	1.69	1.69	1.69	1.69	1.69	1.69	1.69	1.70	1.70	1.70	1.70	1.70
68°F (20.0°C)	TC	27188	28349	29400	29587	30842	28783	28848	29432	29619	30873	30027	30050	30104	30130	30898	32354	32414	32465	32470	32487
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	1.76	1.77	1.78	1.78	1.79	1.77	1.77	1.78	1.78	1.79	1.78	1.78	1.78	1.78	1.79	1.81	1.81	1.81	1.81	1.80
77°F (25.0°C)	TC	26151	27410	28483	28673	29954	27723	27811	28514	28705	29984	28979	29012	29086	29118	30009	31299	31394	31466	31474	31503
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	1.85	1.86	1.87	1.87	1.88	1.87	1.87	1.87	1.87	1.88	1.88	1.88	1.88	1.88	1.88	1.90	1.90	1.90	1.90	1.89
82°F (27.8°C)	TC	25703	27015	28105	28296	29591	27238	27343	28130	28327	29622	28505	28540	28629	28665	29648	30819	30935	31016	31024	31058
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	1.93	1.94	1.95	1.95	1.96	1.95	1.95	1.95	1.95	1.96	1.96	1.96	1.96	1.96	1.96	1.98	1.98	1.98	1.98	1.97
86°F (30.0°C)	TC	25590	26867	27966	28161	29473	26987	27111	27997	28193	29503	28265	28298	28402	28443	29529	30601	30719	30792	30799	30828
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	2.03	2.04	2.05	2.05	2.07	2.04	2.05	2.05	2.05	2.07	2.06	2.06	2.06	2.06	2.07	2.08	2.08	2.08	2.08	2.08
95°F (35.0°C)	TC	25141	25277	26094																	

Climate 5000 Series - Gen 4

24K Btu Single Zone Max Performance System - Wall Mounted IDU



Heating Performance - Indoor Dry Bulb Temperature at 718 CFM										
Outdoor DB(°F)	TCITotal Heating Capacity(BTU/hr)					PIPower Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	13878	13600	13532	13478	13357	2.51	2.53	2.53	2.54	2.55
-13°F(-25.0°C)	16855	16518	16436	16370	16223	2.69	2.71	2.72	2.72	2.73
-4°F(-20.0°C)	21005	20585	20482	20400	20217	2.75	2.77	2.78	2.78	2.79
0°F(-17.8°C)	22977	22518	22405	22315	22115	2.86	2.88	2.89	2.89	2.90
5°F(-15.0°C)	25442	24933	24809	24709	24487	2.99	3.02	3.02	3.03	3.04
14°F(-10.0°C)	26009	25489	25361	25260	25033	3.04	3.07	3.07	3.08	3.09
17°F(-8.3°C)	26198	25674	25546	25443	25214	3.06	3.08	3.09	3.10	3.11
22°F(-5.6°C)	27520	26969	26835	26727	26487	3.06	3.08	3.09	3.10	3.11
27°F(-2.8°C)	28842	28265	28124	28011	27759	3.06	3.08	3.09	3.10	3.11
32°F(0.0°C)	30164	29560	29413	29295	29031	3.06	3.08	3.09	3.10	3.11
35°F(1.7°C)	30957	30338	30186	30065	29795	3.06	3.08	3.09	3.10	3.11
37°F(2.8°C)	30876	30258	30107	29986	29717	2.95	2.98	2.98	2.99	3.00
42°F(5.6°C)	30673	30060	29909	29790	29522	2.68	2.71	2.71	2.72	2.73
44.6°F(7.0°C)	30592	29980	29830	29711	29444	2.58	2.60	2.60	2.61	2.62
47°F(8.3°C)	30470	29861	29712	29593	29327	2.42	2.44	2.44	2.45	2.46
52°F(11.1°C)	29873	29276	29129	29013	28752	2.23	2.25	2.26	2.26	2.27
57°F(13.9°C)	29276	28690	28547	28433	28177	2.05	2.07	2.07	2.08	2.09
59°F(15.0°C)	29037	28456	28314	28201	27947	1.98	2.00	2.00	2.00	2.01
62°F(16.7°C)	28679	28105	27965	27853	27602	1.87	1.89	1.89	1.89	1.90
64.4°F(18.0°C)	28392	27824	27685	27574	27326	1.78	1.80	1.80	1.81	1.81
69.8°F(21.0°C)	27747	27192	27056	26948	26705	1.59	1.60	1.60	1.61	1.61
75.2°F(24.0°C)	27102	26560	26427	26321	26084	1.39	1.40	1.41	1.41	1.41

Bolded values were tested. All other values are simulated and for reference only.